

## NORTHEAST PHILADELPHIA, PA, USA

WMO: 724085

Lat: 40.079N

Lon: 75.013W

Elev: 32

StdP: 100.94

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94732

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |            |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |            | 1%          |             |                       |            |              |
|               | 99.6%        | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-10.7 | (c)<br>-8.2 | (d)<br>-20.4                  | (e)<br>0.6 | (f)<br>-9.2 | (g)<br>-17.9 | (h)<br>0.8 | (i)<br>-6.6 | (j)<br>11.9           | (k)<br>0.1 | (l)<br>11.0 | (m)<br>-0.3 | (n)<br>4.5            | (o)<br>290 | (p)<br>0.481 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 34.0            | 23.9 | 32.6 | 23.2 | 31.3 | 22.6 | 25.6                | 31.5 | 24.7 | 30.1 | 24.0 | 28.9 | 4.3                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.9                            | 18.8 | 28.2 | 23.0 | 17.8 | 27.1 | 22.5 | 17.2 | 26.8 | 78.4          | 31.5 | 74.8 | 30.4 | 72.1 | 29.1 | 30.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.1              | 8.7  | 7.8 | -13.6                      | 36.5 | 2.7                | 1.8 | -15.5                                              | 37.8 | -17.1      | 38.9 | -18.7      | 39.9 | -20.6      | 41.2 |
|                   |      |     | -14.6                      | 27.2 | 2.6                | 1.6 | -16.5                                              | 28.4 | -18.1      | 29.3 | -19.6      | 30.3 | -21.5      | 31.5 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.2   | 0.6   | 2.1   | 6.1   | 12.3  | 17.5  | 22.7  | 25.5  | 24.5  | 20.7  | 14.2  | 8.2   | 3.7   |  |  |
|                                                                        | DBStd       | 9.69   | 5.61  | 5.14  | 4.98  | 4.69  | 4.24  | 3.64  | 2.89  | 2.89  | 3.71  | 4.42  | 4.57  | 4.95  |  |  |
|                                                                        | HDD10.0     | 993    | 295   | 227   | 141   | 26    | 1     | 0     | 0     | 0     | 0     | 10    | 87    | 206   |  |  |
|                                                                        | HDD18.3     | 2571   | 549   | 456   | 379   | 191   | 68    | 7     | 0     | 1     | 18    | 144   | 304   | 455   |  |  |
|                                                                        | CDD10.0     | 2175   | 4     | 5     | 21    | 94    | 234   | 380   | 482   | 451   | 322   | 139   | 33    | 9     |  |  |
|                                                                        | CDD18.3     | 711    | 0     | 0     | 1     | 9     | 43    | 137   | 224   | 193   | 89    | 14    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 6651   | 0     | 1     | 7     | 102   | 424   | 1278  | 2296  | 1767  | 677   | 98    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 2470   | 0     | 0     | 1     | 26    | 128   | 463   | 977   | 658   | 200   | 17    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.5    | 3.9   | 4.0   | 4.3   | 4.0   | 3.4   | 3.2   | 2.9   | 2.7   | 3.0   | 3.2   | 3.5   | 3.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1181   | 94    | 68    | 107   | 98    | 89    | 105   | 113   | 121   | 102   | 96    | 83    | 105   |  |  |
|                                                                        | PrecMax     | 1578   | 151   | 137   | 190   | 261   | 188   | 233   | 272   | 342   | 248   | 243   | 144   | 255   |  |  |
|                                                                        | PrecMin     | 934    | 41    | 17    | 32    | 31    | 24    | 27    | 30    | 36    | 12    | 14    | 27    | 21    |  |  |
|                                                                        | PrecStd     | 166    | 32    | 31    | 41    | 47    | 35    | 53    | 50    | 64    | 62    | 56    | 32    | 48    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 18.5  | 19.9  | 24.5  | 29.8  | 32.1  | 34.5  | 36.5  | 35.5  | 33.5  | 28.6  | 22.7  | 20.3  |  |  |
|                                                                        |             | MCWB   | 15.6  | 15.0  | 16.0  | 18.0  | 22.0  | 23.9  | 25.2  | 24.9  | 23.2  | 21.3  | 17.4  | 16.4  |  |  |
|                                                                        | 2%          | DB     | 15.6  | 16.1  | 20.3  | 26.4  | 30.1  | 32.7  | 34.5  | 33.2  | 31.0  | 26.1  | 20.3  | 17.0  |  |  |
|                                                                        |             | MCWB   | 13.0  | 11.7  | 13.7  | 17.2  | 20.9  | 23.1  | 24.1  | 23.6  | 21.7  | 19.8  | 15.5  | 14.0  |  |  |
|                                                                        | 5%          | DB     | 11.8  | 12.7  | 17.4  | 23.4  | 27.8  | 31.1  | 32.9  | 31.8  | 28.9  | 23.8  | 17.9  | 13.9  |  |  |
|                                                                        |             | MCWB   | 9.0   | 8.3   | 11.8  | 15.6  | 20.0  | 22.5  | 23.6  | 22.9  | 21.0  | 18.4  | 14.1  | 11.2  |  |  |
|                                                                        | 10%         | DB     | 8.6   | 9.8   | 14.5  | 20.9  | 25.8  | 29.3  | 31.5  | 30.2  | 27.3  | 21.8  | 16.0  | 11.2  |  |  |
|                                                                        |             | MCWB   | 6.1   | 6.4   | 9.7   | 14.3  | 18.6  | 21.3  | 22.9  | 22.1  | 20.4  | 17.3  | 12.4  | 8.3   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 16.7  | 16.3  | 17.7  | 20.4  | 23.5  | 25.6  | 27.6  | 26.9  | 25.2  | 23.0  | 18.8  | 17.4  |  |  |
|                                                                        |             | MCDB   | 17.9  | 20.1  | 23.0  | 25.9  | 29.7  | 31.1  | 32.8  | 32.4  | 30.6  | 25.9  | 21.3  | 19.0  |  |  |
|                                                                        | 2%          | WB     | 13.3  | 12.6  | 14.9  | 18.5  | 22.1  | 24.4  | 25.7  | 25.3  | 23.9  | 21.2  | 17.0  | 14.5  |  |  |
|                                                                        |             | MCDB   | 15.3  | 15.2  | 18.9  | 24.1  | 28.1  | 30.5  | 32.0  | 30.9  | 27.7  | 24.8  | 19.1  | 16.5  |  |  |
|                                                                        | 5%          | WB     | 9.1   | 9.1   | 12.6  | 16.9  | 20.9  | 23.7  | 24.9  | 24.4  | 22.9  | 19.6  | 15.0  | 11.8  |  |  |
|                                                                        |             | MCDB   | 10.9  | 11.8  | 16.2  | 21.9  | 26.2  | 29.2  | 30.6  | 29.2  | 26.5  | 22.8  | 17.3  | 13.8  |  |  |
|                                                                        | 10%         | WB     | 6.5   | 6.6   | 10.0  | 15.2  | 19.6  | 22.8  | 24.1  | 23.7  | 21.9  | 17.8  | 13.0  | 8.7   |  |  |
|                                                                        |             | MCDB   | 8.4   | 9.4   | 14.3  | 20.1  | 24.3  | 27.7  | 29.2  | 28.0  | 25.7  | 20.7  | 15.4  | 10.7  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.9   | 8.6   | 9.7   | 10.9  | 10.5  | 10.1  | 9.8   | 9.5   | 9.9   | 9.9   | 9.3   | 7.8   |  |  |
|                                                                        | 5% DB       | MCDBR  | 11.9  | 13.0  | 14.3  | 14.5  | 13.0  | 11.5  | 11.2  | 10.7  | 11.4  | 11.9  | 11.8  | 11.1  |  |  |
|                                                                        |             | MCWBR  | 10.0  | 9.8   | 9.4   | 7.7   | 6.2   | 5.0   | 4.2   | 4.2   | 4.9   | 6.7   | 8.3   | 9.7   |  |  |
|                                                                        | 5% WB       | MCDBR  | 11.1  | 12.1  | 13.3  | 12.6  | 11.7  | 10.2  | 9.9   | 8.9   | 9.0   | 9.5   | 9.8   | 10.5  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.312 | 0.321 | 0.361 | 0.392 | 0.439 | 0.501 | 0.501 | 0.494 | 0.407 | 0.361 | 0.328 | 0.310 |  |  |
|                                                                        | taud        |        | 2.456 | 2.419 | 2.372 | 2.313 | 2.212 | 2.080 | 2.095 | 2.101 | 2.315 | 2.454 | 2.508 | 2.515 |  |  |
|                                                                        | Ebn at Noon |        | 853   | 896   | 891   | 882   | 844   | 790   | 785   | 778   | 831   | 840   | 828   | 830   |  |  |
|                                                                        | Edh at Noon |        | 79    | 94    | 110   | 125   | 141   | 162   | 158   | 153   | 116   | 91    | 75    | 70    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.91  | 2.82  | 3.73  | 4.83  | 5.37  | 5.88  | 6.00  | 5.20  | 4.32  | 3.07  | 2.18  | 1.66  |  |  |
|                                                                        | RadStd      |        | 0.17  | 0.31  | 0.30  | 0.43  | 0.63  | 0.49  | 0.29  | 0.34  | 0.47  | 0.32  | 0.19  | 0.17  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.62 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -129    | +190    | +119    |
| (47) Regional (72 neighbors) |  | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +122    | +69     |

Nomenclature: See separate page